COMPUTER PROGRAMMING

Name: Talha Ali
Registration No: BF24NWELE0728
Section: A (Power)
Electrical Department
UET Nowshehra

LAB 2

LAB TITLE

Basic Syntax and Input/Output in C++

---

1. OBJECTIVE

The objective of this lab is to understand the basic structure of a C++ program. In this lab we learn how to use input and output statements, work with different data types, apply indentation, and perform calculations using arithmetic operators.

2. THEORY

C++ programs follow a specific structure that includes header files, a main function, and program statements. The iostream library is used for input and output operations.

The cout statement is used to display output on the screen, while cin is used to take input from the user. The symbols << and >> are known as stream insertion and extraction operators.

C++ is a statically typed language, which means every variable must be declared with a data type before it is used. Some common data types include:

- **int for integers**

- **double for decimal numbers**
- **char for single characters**
- **bool for true/false values**
- **string for text**

C++ also provides arithmetic operators such as +, −, *, /, and % which are used to perform mathematical calculations. Proper indentation makes the program easier to read and understand. Comments are used to explain the code and can be written using // for single line comments or / / for multi-line comments.

3. SOFTWARE

- DEV-C++ IDE
- C++ Compiler (included in DEV-C++)

PROGRAM 1

Hello World and User Greeting

Algorithm

1. Start the program
2. Declare a variable to store the user's name
3. Ask the user to enter their name
4. Display a greeting message
5. End the program

```
#include <iostream>
using namespace std;

int main ()
{
   string name;

   cout << "Enter your name: ";
     cin >> name;
  cout << "Hello " << name << "! Welcome to C++ programming.";
  return 0; }
```

PROGRAM 2

Simple Calculator Using Arithmetic Operators

Algorithm

- Start the program
- Declare two numbers
- Take input from the user
- Perform addition, subtraction, multiplication, and division
- Display the results
- End the program

```
#include <iostream> using namespace std;

int main()
{     int a, b;

    cout << "Enter two numbers: ";     cin >> a >> b;

    cout << "Addition = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    cout << "Division = " << a / b << endl;

    return 0;
}
```

Conclusion

In this lab we learned the basic structure of a C++ program. We also learned how to use input and output statements, work with different data types, and perform calculations using arithmetic operators. This lab helped us understand the fundamentals of C++ programming.